

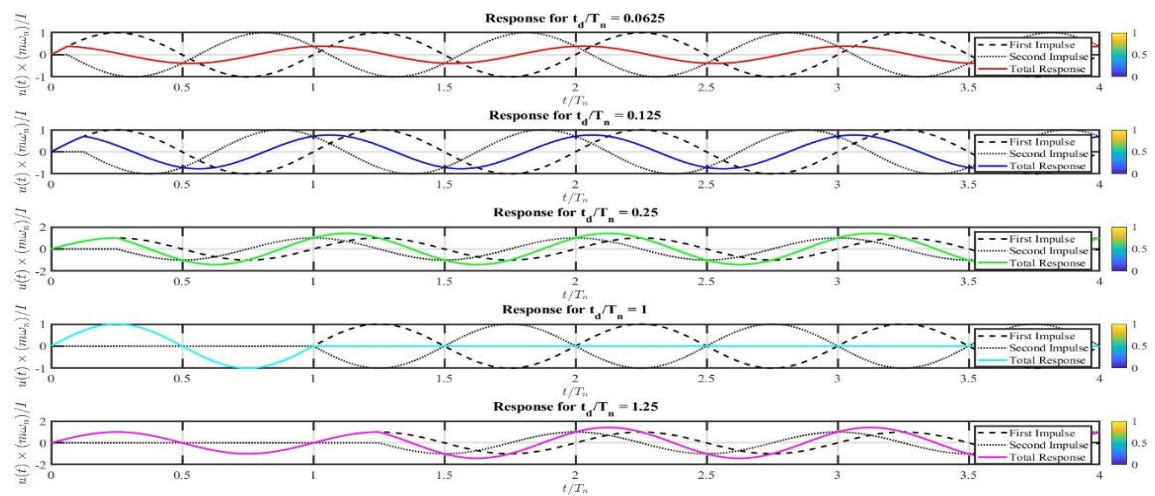
Question 1: SDOF System Subjected to Two Impulses

An undamped SDOF system is subjected to a force $p(t)$ consisting of a sequence of two impulses of magnitude I , as shown below. (a) Plot the normalized displacement $(u(t) * (m\omega_n)/I)$ response of the system for $t_d/T_n = 1/8, 1/4$, and 1 . For each case show the response to individual impulses and the combined response. (b) Plot normalized amplitude $(u_o * (m\omega_n)/I)$ as a function of t_d/T_n . Such a plot is called the response spectrum for this excitation.

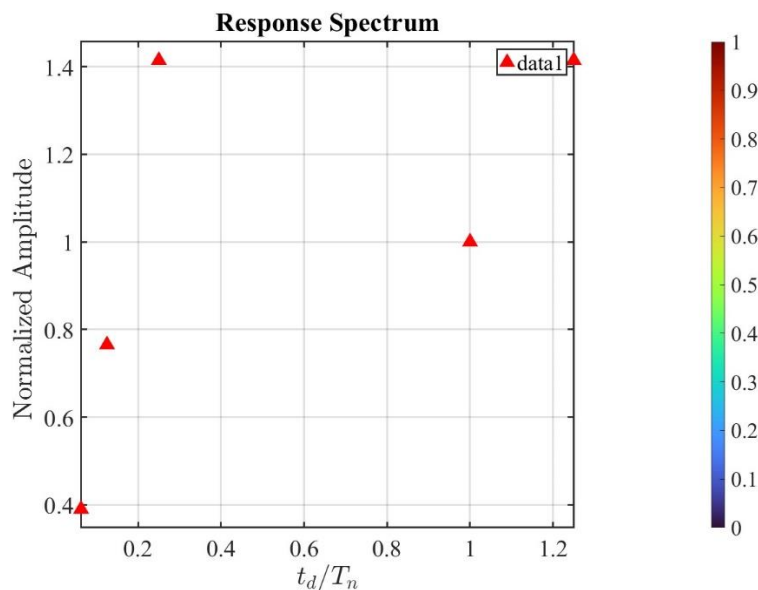
MATLAB

Solution:

- (a) Plotting the normalized displacement $(u(t) * (m\omega_n)/I)$ response of the system for $t_d/T_n = 0.0625, 0.125, 0.25, 1$, and 1.25 . For each case showing the response to individual impulses and the combined response.



- (b) Now, Plotting the normalized amplitude $(u_o * (m\omega_n)/I)$ as a function of t_d/T_n .



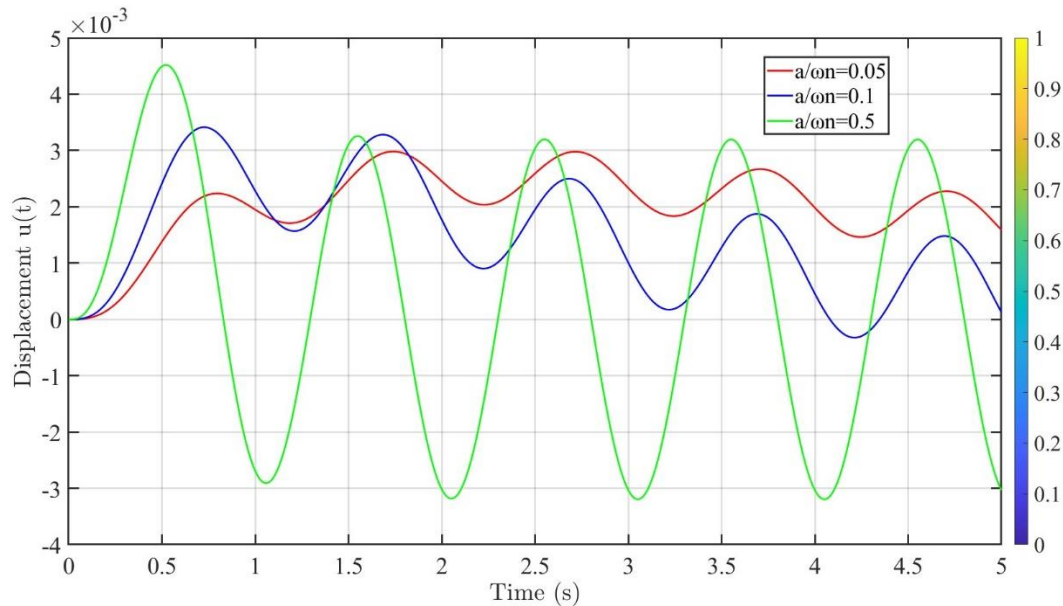
Such plotting is called the response spectrum for this excitation.

Question 2: Undamped System Subjected to Loading p(t)

Derive the expression for the motion of an undamped system subjected to the loading p(t) shown in the figure below. Plot the motion for $b=2a$ and $a/\omega n = 0.05, 0.1$ and 0.5 .

MATLAB

Solution: Now, plotting for the motion for $b=2a$ and $a/\omega n = 0.05, 0.1$ and 0.5 .



Question 3: Full Elevated Water Tank

The full elevated water tank weighs 100 kips. Find the maximum lateral displacement of the tower subjected to each of the two forces shown in the figures below. The tower has a lateral stiffness of 8.2 kips/in. Use your understanding of how the maximum response depends on the ratio of the rise time of the applied force to the natural vibration period of the system neglecting damping.

Solution: Given,
Weight, $W = 100$ kips.
Lateral Stiffness, $k = 8.2$ kips/in.
Natural frequency $\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{8.2 \times 386.4}{100}} \approx 5.62 \text{ rad/s}$.
Natural Period: $T_n = \frac{2\pi}{\omega_n} \approx 1.12 \text{ sec}$.
Static Displacement $u_{st} = \frac{W}{k} = \frac{50}{8.2} \approx 6.10$ inches.
DAF: For a ramp force with rise time t_r , the DAF is
$$DAF = 1 + \frac{1}{\pi r} \sin(\pi r); \text{ where } r = \frac{t_r}{T_n}.$$

Case 1: $t_r = 0.2 \text{ sec}$;
 $r = \frac{0.2}{1.12} \approx 0.1786$;
$$DAF = 1 + \frac{1}{\pi \cdot 0.1786} \cdot \sin(\pi \cdot 0.1786) \approx 1.95$$

Maximum Displacement
$$u_{max} = DAF \cdot u_{st} = 1.95 \times 6.10 \approx \boxed{11.89 \text{ inches}}$$

Case 2+ $t_r = 80 \text{ sec}$

$$r = \frac{80}{1.12} \approx 71.43:$$

$DAF \approx 1$ (Static loading).

Maximum Displacement

$$u_{max} = u_{st} = \boxed{6.10 \text{ inches}}$$

Answers: (a). 11.89 inches; (b). 6.10 inches.